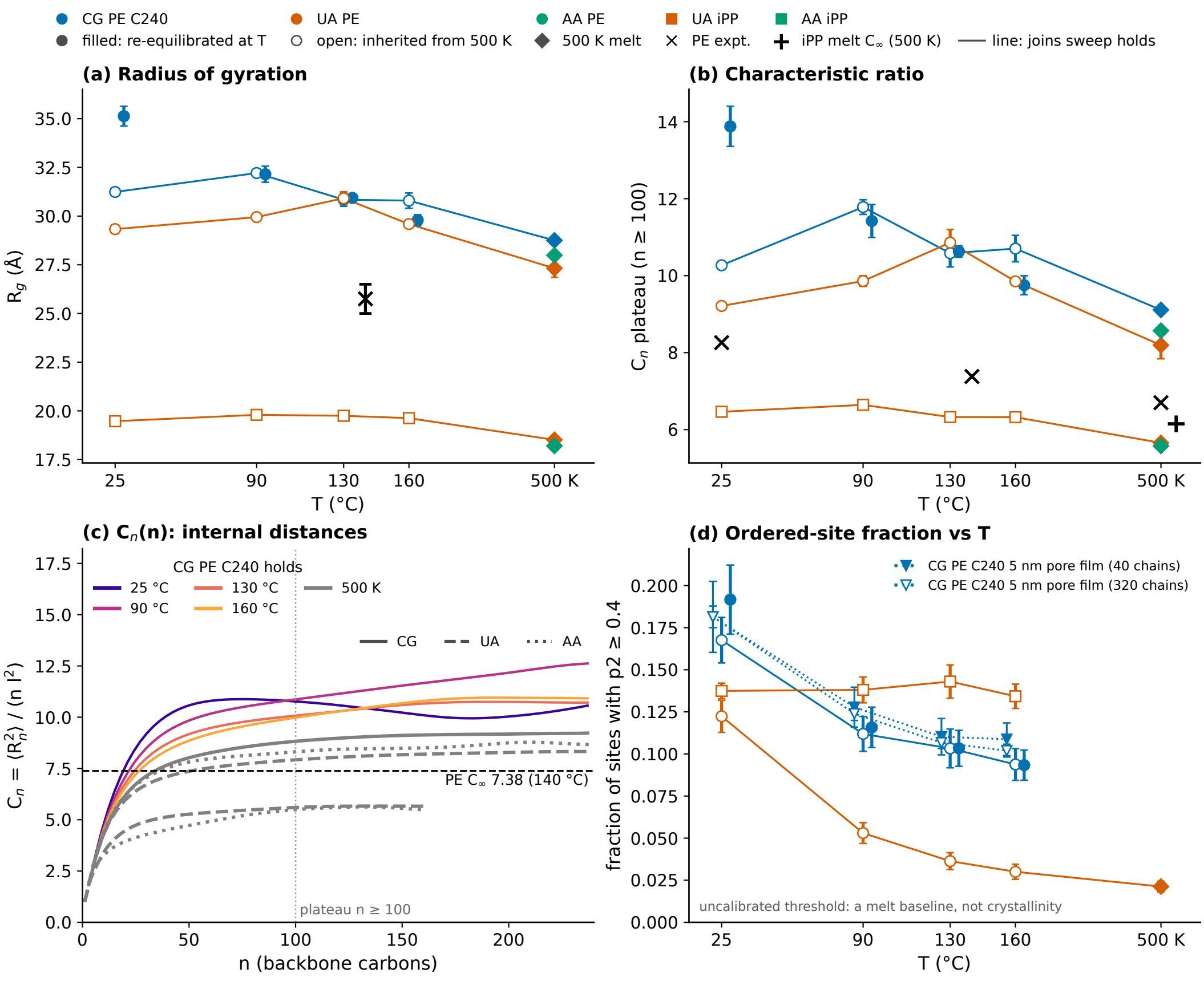
CG PE C240
UA PE
AA PE
UA iPP
AA iPP
filled: re-equilibrated at T
open: inherited from 500 K
500 K melt
PE expt.
iPP melt C∞ (500 K)
line: joins sweep holds
(a) Radius of gyration
R_g (Å)
T (°C)
(b) Characteristic ratio
C_n plateau (n ≥ 100)
T (°C)
(c) C_n(n): internal distances
CG PE C240 holds
25 °C
90 °C
130 °C
160 °C
500 K
CG
UA
AA
PE C∞ 7.38 (140 °C)
plateau n ≥ 100
C_n = ⟨R_n²⟩ / (n l²)
n (backbone carbons)
(d) Ordered-site fraction vs T
CG PE C240 5 nm pore film (40 chains)
CG PE C240 5 nm pore film (320 chains)
fraction of sites with p2 ≥ 0.4
uncalibrated threshold: a melt baseline, not crystallinity
T (°C)